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DELIVERABLE

D6.8 – Procedural Ocean Read Me Documentation

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Executive Summary

This document is the read me file of the Procedural Ocean platform, providing detailed explanations of different components needed to run the application.

For visual reference please see the following two videos:

[Procedural Ocean \(youtube.com\)](#)

Video 1: Procedural Ocean Trailer

[Immersive Ocean Digital Twin Using Procedural Content Generation in Unreal Game Engine \(youtube.com\)](#)

Video 2: Webinar on using procedural content generation for Procedural Ocean

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Document History			
Version	Date	Contributor(s)	Description
1.0	09/07/2024	BUas	Finished PC & VR build of PO

Definitions, Acronyms and Abbreviations

Acronym/ Abbreviation	Title
BUas	Breda University of applied sciences
UI	User interface
PO	Procedural Ocean

1 Copyright and Licence

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2 How to Contribute

The open-source version of Procedural Ocean can be found on the following GIT link:
<https://github.com/BredaUniversityResearch/MSP-ProceduralOceanView>

(This version omits certain paid Unreal assets, as they could not be made open-source at the time of delivery. However, when utilizing Procedural Rule Packs, it is possible to incorporate the newly added assets into the original version of the PO).

Within Unreal Engine, we have developed an asset named Procedural Rule Packs (PRPs). These PRPs are organizational content base folders that contain specific procedural rules for generating and coordinating the behavior of related assets. In the Documentation folder (located next to this Read Me file), you will find a PDF titled "Procedural Rule Pack Guide." This document provides detailed instructions on how to add a PRP into Unreal to ensure compatibility with PO.

3 Import Custom JSON Data

Launch the **pov.launcher.exe.** application to start adding custom JSON data. Select option '**1. Import POV config**' to initiate the importing sequence. You will be prompted to enter the coordinates of the region you wish to import. To obtain these coordinates, run the '**pov-launcher.ps1**' script, which allows you to search for '**Region coordinate system: <https://epsg.io/3035>**.' There, you will find a button labeled '**Get position on the map,**' which will copy the coordinates for you. Enter these coordinates into **pov.launcher.exe.** Leave the other values at their default settings by pressing Enter.

4 Set Up Unreal Engine

Ensure that your PC meets the requirements to run a game engine. You can find the necessary specifications here: <https://catnessgames.com/blog/unreal-engine-5-requirements/>

&

https://dev.epicgames.com/documentation/en-us/unreal-engine/hardware-and-software-specifications-for-unreal-engine?application_version=5.3

Download and install the "**Epic Game Launcher**": <https://store.epicgames.com/en-US/download>. In your "**Epic Games Launcher**", make sure you have the latest version of Unreal 5.3 installed (5.3.x). If you don't, select the "+" icon next to "**Engine Versions**", and install the correct version. Once Unreal is installed you can launch the project via the launcher, or from the Git folder.

5 Create Builds

To create builds in Unreal Engine for a playable execute, you can follow the steps in the following link: https://dev.epicgames.com/documentation/en-us/unreal-engine/preparing-unreal-engine-projects-for-release?application_version=5.3

(This version is missing certain paid Unreal assets, as they could not be made open-source at the time of delivery. Consequently, some components may be absent in your new build compared to the original PO build).

6 Virtual Reality Mode

Connect your Meta Quest 3 headset to the PC. Then, launch the **pov.launcher.exe** application and select option '**4. Launch in VR mode**'. You will be prompted to select a JSON file; choose the desired file and press Enter.

Disclaimer: The weather controls + minimap are disabled in VR mode to safe performance, ensuring a better player experience.

General Controls:

- **Moving through world:** By **pointing** the controller of choice in a specific direction and pressing the **grip button(s)** on the handle.
- **Adjust movement speed:** By swiping up- or downwards on a **trackpad** (Vive) or pressing up or down on a **joystick** (Quest 3) on one of the controllers.
- **Interaction with UI and other world elements:** By **pointing** at an element with one of the controllers and pressing the **trigger** on the back to interact with it.
- **Snapturn -/+ 45 degrees:** By swiping left or right on a **trackpad** (Vive) or pressing left or right on a **joystick** (Quest 3) on one of the controllers.
- **Open menu:** By **rotating your hand** of choice sideways, **as if reading your watch** (advice to use recessive hand).
*(Additionally, we could also implement the use of the **menu button** on the left-hand controller. A first press would open the menu, a second press would close it.)*

UI Interaction:

1. Take a controller in each hand.
2. Open the menu by **rotating your hand** of choice sideways, **as if reading your watch** (advice to use recessive hand). The menu will open above said controller. The menu will remain open as long as you keep your hand in the correct position, even if the UI moves out of your field of view.
*(Additionally, we could also implement the use of the **menu button** on the left-hand controller. A first press would open the menu, a second press would close it.)*
3. Select one of the menu icons by **pointing** the second controller and pressing the **trigger** on the back (advice to use dominant hand).
4. **Data layers window:** Select one of the data layers from the list through point and click, as described above. This will open a second window on its right side which contains all the available display modes for said data layer.
5. **Specific data layer display modes window (example: wind farms):** Toggle on one or multiple display modes of interest through point and click. This will visualise the data in the world.

When applicable, for example in the case of a **heat map**, a display mode will be accompanied by a legend.

When applicable, for example in the case of **area points** (wind turbine models), a display mode will be accompanied by a button that allows the user to cycle through the various phases. Each phase will visualise the model in a different state of construction. (In the future, this feature will be linked to the timeline.)

Each display mode can be separately collapsed or expanded through point and click.

Loss of Tracking:

If a controller loses tracking, an animated loading circle should appear (in the middle of the screen?) with a warning 'Reconnecting controller x'.